OS ORANGE ASSIGNMENT

Name: Kaushik Balaji

SRN: PES2UG24CS220

Semester: 4 **Section:** D

Github Repo link: https://github.com/KaushikBalaji05/PES2UG24CS220-pes-vcs

Phase1:

A)

```
kaushik@kaushik-VirtualBox:~$ cd PES2UG24CS220-pes-vcs
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

B)

```
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$
```

Phase 2:

A)

```
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ make test_tree
make: 'test_tree' is up to date.
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$
```

B)

```
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ find .pes/objects -type f
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
```

Phase 3:

A)

```
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ echo "hello" > file1.txt
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ echo "world" > file2.txt
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ ./pes add file1.txt file2.txt
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  index.c
  untracked:  test_sequence.sh
  untracked:  index.h
  untracked:  pes.o
  untracked:  object.o
  untracked:  tree.c
  untracked:  pes.c
  untracked:  Makefile
  untracked:  test_tree.c
  untracked:  pes.h
  untracked:  test_objects.c
  untracked:  commit.h
  untracked:  commit.o
  untracked:  tree.o
  untracked:  a.txt
  untracked:  b.txt
  untracked:  README.md
  untracked:  index.o
  untracked:  commit.c
  untracked:  tree.h
  untracked:  object.c
  untracked:  src
```

B)

```
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776426035 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776426039 6 file2.txt
```

Phase 4:

A)

```
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ echo "Hello" > hello.txt
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ ./pes add hello.txt
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ ./pes commit -m "Initial Commit"
Committed: c70a1da1050a... Initial Commit
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ echo "World" >> hello.txt
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ ./pes add hello.txt
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ ./pes commit -m "Add World"
Committed: e1cbb5486f4c... Add World
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ echo "Goodbye" > bye.txt
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ ./pes add bye.txt
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ ./pes commit -m "Add Farewell"
Committed: afe4a74afad1... Add Farewell
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ ./pes log
commit afe4a74afad171f3df659674100aa5a655f07c7892e8207f2bf16fa327bbbbb0
Author: PES User <pes@localhost>
Date:   1776426246

    Add Farewell

commit e1cbb5486f4c92ba531a4504aa5234ceead96f1a22839e17167ca4bc18322038
Author: PES User <pes@localhost>
Date:   1776426218

    Add World

commit c70a1da1050a1b44133ce2075fa68a0b098d23b715e99071d512e27ea5e96faf
Author: PES User <pes@localhost>
Date:   1776426187

    Initial Commit
```

B)

```
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/af/e4a74afad171f3df659674100aa5a655f07c7892e8207f2bf16fa327bbbbb0
.pes/objects/c7/0a1da1050a1b44133ce2075fa68a0b098d23b715e99071d512e27ea5e96faf
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e1/cbb5486f4c92ba531a4504aa5234ceead96f1a22839e17167ca4bc18322038
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/refs/heads/main
```

C)

```
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ cat .pes/refs/heads/main
afe4a74afad171f3df659674100aa5a655f07c7892e8207f2bf16fa327bbbbb0
kaushik@kaushik-VirtualBox:~/PES2UG24CS220-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

Phase 5 and 6 (Analysis-Only Questions):

Q5.1) pes checkout <branch> updates .pes/HEAD to point to the branch and reads the commit hash from .pes/refs/heads/<branch>. The working directory is then updated to match the commit's tree by adding, modifying, or deleting files. The operation is complex because it must safely update files without overwriting uncommitted changes.

Q5.2) Compare working directory files with the index:

- missing file → deleted
- different size/mtime or hash → modified

If a file is modified locally and differs in the target branch, checkout must refuse. This detects a dirty working directory.

Q5.3) In detached HEAD, commits are not linked to any branch. They become unreachable if you switch branches. To recover, create a branch at that commit or use reflog to find it.

Q6.1) Start from all branch heads and traverse commits, trees, and blobs, marking reachable objects using a hash set. Delete objects not in the set. For 100k commits, about 100k objects (plus trees/blobs) are visited once.

Q6.2) GC can delete objects that a commit is about to reference if run concurrently. Git avoids this using atomic writes, updating objects before references, and protecting recent objects from deletion.